

5.3.28 USB_FS characteristics

Table 71. USB_FS characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V _{DDUSB}	USB transceiver operating supply voltage	-	3.0 ⁽¹⁾	-	3.6	V
R _{PUI}	Embedded USB_DP pullup value during idle	-	900	-	1575	Ω
R _{PUR}	Embedded USB_DP pullup value during reception	-	1425	-	3090	
Z _{DRV}	Output driver impedance ⁽²⁾	High and low driver	28	36	44	

1. USB functionality is ensured down to 2.7 V, but some USB electrical characteristics are degraded in 2.7 to 3.0 V range.
2. No external termination series resistors are required on USB_DP (D+) and USB_DM (D-). The matching impedance is already included in the embedded driver.

5.3.29 Ethernet interface characteristics

Unless otherwise specified, the parameters given in the following Tables are derived from tests performed under the ambient temperature, frcc_c_ck frequency, and V_{DD} supply voltage conditions summarized in [Table 18. General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- Capacitive load Cl = 20 pF
- Measurements points are done at CMOS levels : 0.5 V_{DD}
- IO compensation cell activated

Table 72. Dynamic characteristics: Ethernet MAC signals for SMI

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
t _{MDC}	MDC cycle time(2.5 MHz)	399	400	401	ns
T _d (MDIO)	Write data valid time	0.5	1	1.5	
t _{su} (MDIO)	Read data setup time	14.5	-	-	
t _h (MDIO)	Read data hold time	0	-	-	

Table 73. Dynamic characteristics: Ethernet MAC signals for RMII

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
t _{su} (RXD)	Receive data setup time	2	-	-	ns
t _{ih} (RXD)	Receive data hold time	1	-	-	
t _{su} (CRS)	Carrier sense setup time	1.5	-	-	
t _{ih} (CRS)	Carrier sense hold time	1	-	-	
t _d (TXEN)	Transmit enable valid delay time	7.5	9.5	14	
t _d (TXD)	Transmit data valid delay time	7.5	10	15.5	

Table 74. Dynamic characteristics: Ethernet MAC signals for MII

Symbol	Parameter	Min	Typ	Max	Unit
t _{su} (RXD)	Receive data setup time	1.5	-	-	ns
t _{ih} (RXD)	Receive data hold time	1.5	-	-	
t _{su} (DV)	Data valid setup time	1.5	-	-	